

# Pronab Kumar Paul

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Computer Vision Researcher — Medical AI

*Computer Vision and Healthcare AI researcher with peer-reviewed publications in Springer proceedings and engineering journals. Research focuses on explainable AI, robust keypoint descriptors, 3D vision, and federated learning for medical image analysis. Currently developing robust and anatomy-aware visual representations for medical imaging and intelligent vision systems.*

## RESEARCH INTERESTS

Computer Vision and Medical Image Analysis; Explainable and Anatomy-Aware AI (XAI); Classical and Deep Keypoint Descriptors and Feature Matching; 3D Vision and Geometric Reasoning (6-DoF Pose Estimation); Federated and Privacy-Preserving Learning; Vision-Centric IoT and Robotics Systems.

## EDUCATION

University of Rajshahi, Bangladesh

Jan 2020 – Oct 2025

B.Sc. in Information and Communication Engineering

GPA: 3.82/4.00

Rank: 4/65 (Top 6%)

Honors: Dean's Award (2020–2025) for sustained academic excellence

**Relevant Coursework:** Artificial Intelligence and Neural Computing, Digital Image Processing, Probability & Statistics, Linear Algebra, Programming with C, Data Structures and Algorithms, Java and Network programming, Cryptography, Information Theory and Coding, Database Management System.

## TECHNICAL SKILLS

**Programming & Frameworks:** Python, C/C++, Java, JavaScript, SQL, MATLAB, PyTorch, TensorFlow, Keras, Scikit-learn, Docker, Git

Design, Geometric Transformations, Epipolar Geometry, Camera Projection Models

**Computer Vision & AI:** Classical and deep keypoint descriptors (SIFT, ORB, BRISK, SuperPoint, R2D2, D2-Net, DISK, Key.Net+HyNet), CNNs, Vision Transformers (ViT), CNN-Transformer hybrids, RANSAC, Structure-from-Motion (SfM), Multi-View Stereo, 6-DoF Pose Estimation

**Algorithms & Modeling:** Feature Matching Pipelines, Optimization Algorithms (Gradient-Based, RANSAC Variants), Graph-Based Matching, Algorithm Design and Analysis, Pseudo-Code and Model Derivation

**Statistical Analysis & Evaluation:** Hypothesis Testing, Confidence Intervals, Statistical Significance, Robust Error Metrics, Performance Benchmarking, Ablation Studies, Cross-Validation

**Medical AI & Explainable AI (XAI):** OCT/Image Registration and Evaluation, Lesion Localization, Anatomy-Aware Spatial Constraints, SHAP, LIME

**Mathematical Foundations:** Linear Algebra, Probability Theory, Optimization Methods, Loss Function

**IoT & Engineering:** ML-Based Antenna Design and Optimization, CST Studio Suite, Arduino, Simulation Automation

**Web Development & Tools:** Node.js, Express.js, REST API, MongoDB, OpenCV, VS Code, LaTeX

## RESEARCH EXPERIENCE

Computer Vision Lab, University of Rajshahi

2024 – Present

Research Assistant — Supervisor: Dr. Dipankar Das

Focus: Developing robust computer vision and healthcare AI pipelines.

- Conducted large-scale benchmarking of 17+ classical keypoint detectors-descriptors on real-world imaging scenarios (**Under Review, SSRN**).
- Designed and implementing a Federated Knowledge Distillation framework for multi-institutional medical imaging, enabling privacy-preserving collaborative learning.
- Leading a research team to develop anatomy-aware AI models with explainable visual outputs.

Antenna & Communication Lab, University of Rajshahi

May 2025 – Sep 2025

Research Intern — Supervisor: Dr. Md. Firoz Ahmed

Focus: ML-driven optimization of antenna design for IoT applications.

- Developed ML-based surrogate models for patch antenna performance prediction, achieving  $R^2 \approx 0.95$  and reducing CST simulation time by **60%** through an automated design framework.
- Contributed to 5 journal papers on wireless sensor networks and IoT applications.

SELECTED PUBLICATIONS

[1] **Paul, P.K.**, et al. "Handling Imbalanced Datasets with Real-World Positive Samples for Dengue Prediction Using Machine and Deep Learning Models." *Communications in Computer and Information Science, Springer*, 2025. [doi:10.1007/978-3-032-11335-1\\_29](https://doi.org/10.1007/978-3-032-11335-1_29)

[2] Ahmed, M.F., ..., **Paul, P.K.**, et al. "Compact Patch Antenna for Wireless Sensor Networks." *Applied Engineering & Technology*, 2024.

[3] **Paul, P.K.**, et al. Additional 4 journal papers on antenna design and IoT applications (2024–2025).

Submitted / Under Review:

- **Paul, P.K.**, Das, D. "Benchmarking the Robustness of Classical Keypoint Detectors and Descriptors across Real-World Imaging Scenarios." *Under review. SSRN Preprint*, 2025

SELECTED TECHNICAL PROJECTS

Attention-Guided Lesion Localization with Explainable AI 2025  
*Medical Imaging, Explainable AI*

- Proposed descriptor-guided attention mechanism using ResNet-50 for anatomically interpretable lesion localization.
- Generated attention-weighted similarity maps to support transparent clinical decision-making.
- **Technologies:** PyTorch, Attention Mechanisms, Medical Imaging

Comprehensive OCT Image Registration Benchmarking 2025  
*Medical Image Analysis*

- Benchmarked five OCT registration algorithms under real-world distortions with quantitative statistical validation.
- Evaluated using SSIM, PSNR, RMSE, NCC, and mutual information.
- **Technologies:** Python, OpenCV, Statistical Evaluation

Transformer-Based 6-DoF Surgical Tool Pose Estimation 2024  
*Computer Vision, Robotics, Medical AI*

- Designed a hybrid CNN–Transformer architecture for keypoint detection and real-time 6-DoF pose estimation.
- Proposed tensor-aligned rotation loss to stabilize SO(3) convergence under noisy surgical scenes.
- **Technologies:** PyTorch, Vision Transformers, Geometry-Aware Losses

Federated & Explainable AI for Healthcare Prediction 2024  
*Machine Learning, Healthcare AI*

- Built end-to-end XAI pipelines using SHAP and LIME for clinician-facing interpretability.
- Applied federated learning strategies to address data imbalance and privacy constraints.
- **Technologies:** Python, Scikit-learn, SHAP, Federated Learning

Medical Imaging Foundation Models Analysis 2023  
*Vision Foundation Models, Retinal Imaging*

- Compared DINOv2 and OCT-Transformer with CNN baselines for retinal disease detection and segmentation.
- Achieved an average performance of 0.939.
- **Technologies:** PyTorch, Vision Foundation Models, Medical Imaging

Scalable Photography Contest Platform 2022  
*Full-Stack Systems — Team Lead*

- Led full-stack development of a secure platform for artwork submission, exhibition, and voting for 500+ concurrent users.
- **Technologies:** Node.js, Express.js, MongoDB, JavaScript

AWARDS & HONORS

- **General Merit Scholarship** (2025–2026)  
University of Rajshahi – awarded for academic excellence.
- **Dean’s Award** (2020–2025)  
University of Rajshahi – for sustained academic excellence.
- **Academic Rank: 4th out of 65 (Top 6%)**  
B.Sc. in Information & Communication Engineering.
- **Second Runner-Up, Learnathon 3.0** – developed a custom gRPC-based code execution engine.

REFERENCES

Available upon request.